

Table 623. Cortex-M7 PPB ROM table register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFE0	M7_PPBBROM_PIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
	Reset value																									1	1	0	0	0	1	1	1
0xFE4	M7_PPBBROM_PIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
	Reset value																									1	0	1	1	0	1	0	0
0xFE8	M7_PPBBROM_PIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC		JEP106ID[6:4]	
	Reset value																									0	0	0	0	1	0	1	1
0xFEC	M7_PPBBROM_PIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
	Reset value																									0	0	0	0	0	0	0	0
0xFF0	M7_PPBBROM_CIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
	Reset value																									0	0	0	0	1	1	0	1
0xFF4	M7_PPBBROM_CIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
	Reset value																									0	0	0	1	0	0	0	0
0xFF8	M7_PPBBROM_CIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
	Reset value																									0	0	0	0	0	1	0	1
0xFFC	M7_PPBBROM_CIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
	Reset value																									1	0	1	1	0	0	0	1

60.6.2 Cortex-M7 data watchpoint and trace unit (DWT)

The DWT provides four comparators that can be used as:

- watchpoint
- ETM trigger
- PC sampling trigger
- data address sampling trigger
- data comparator (comparator 1 only)
- clock cycle counter comparator (comparator 0 only)

It also contains counters for:

- clock cycles
- folded instructions
- load store unit (LSU) operations
- sleep cycles
- number of cycles per instruction
- interrupt overhead

A DWT comparator compares one of the following with the value held in its DWT_COMP register:

- a data address
- an instruction address
- a data value
- the cycle count value, for comparator 0 only.

For address matching, the comparator can use a mask, so it matches a range of addresses.

On a successful match, the comparator generates one of the following:

- one or more DWT data trace packets, containing one or more of:
 - the address of the instruction that caused a data access
 - an address offset, bits[15:0] of the data access address
 - the matched data value
- a watchpoint debug event, on either the PC value or the accessed data address
- a CMPMATCH[N] event that signals the match outside the DWT unit

A watchpoint debug event either generates a DebugMonitor exception, or causes the processor to halt execution and enter Debug state.

For more details on how to use the DWT, refer to the Arm®v7-M Architecture Reference Manual [5].

Cortex-M7 DWT registers

DWT control register (M7_DWT_CTRL)

Address offset: 0x000

Reset value: 0x4000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NUMCOMP[3:0]				NOTRCPKT	NOEXTTRIG	NOCYCCNT	NOPRFCNT	Res.	CYCEVTENA	FOLDVTENA	LSUEVTENA	SLEEPEVTENA	EXCEVTENA	CPIEVTENA	EXCTRCENA
r	r	r	r	r	r	r	r		rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	PCSA MPLE NA	SYNCTAP[1:0]		CYCTAP	POSTINIT[3:0]				POSTRESET[3:0]				CYCCNTENA
			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 **NUMCOMP[3:0]**: Number of comparators implemented (read-only)

0x4: Four comparators

Bit 27 **NOTRCPKT**: Trace sampling and exception tracing support (read-only)

0: Supported

Bit 26 **NOEXTTRIG**: External match signal, CMPMATCH support (read-only)

0: Supported

Bit 25 **NOCYCCNT**: Cycle counter support (read-only)

0: Supported

- Bit 24 **NOPRFCNT**: Profiling counter support (read-only)
0: Supported
- Bit 23 Reserved, must be kept at reset value.
- Bit 22 **CYCEVTENA**: POSTCNT underflow event counter packet generation enable
0: Disabled
1: Enabled
- Bit 21 **FOLDEVTEA**: Folded instruction counter overflow event generation enable
0: Disabled
1: Enabled
- Bit 20 **LSUEVTENA**: LSU counter overflow event generation enable
0: Disabled
1: Enabled
- Bit 19 **SLEEPEVTENA**: Sleep counter overflow event generation enable
0: Disabled
1: Enabled
- Bit 18 **EXCEVTENA**: Exception overhead counter overflow event generation enable
0: Disabled
1: Enabled
- Bit 17 **CPIEVTENA**: CPI counter overflow event generation enable
0: Disabled
1: Enabled
- Bit 16 **EXCTRCENA**: Exception trace generation enable
0: Disabled
1: Enabled
- Bits 15:13 Reserved, must be kept at reset value.
- Bit 12 **PCSAMPLENA**: POSTCNT counter use enable
Enables use of POSTCNT counter as a timer for Periodic PC sample packet generation.

0: Disabled
1: Enabled
- Bits 11:10 **SYNCTAP[1:0]**: Position of synchronization packet counter tap on CYCCNT counter
This selection determines the synchronization packet rate.

0x0: Disabled - no synchronization packets
0x1: Tap at CYCCNT[24]
0x2: Tap at CYCCNT[26]
0x3: Tap at CYCCNT[28]
- Bit 9 **CYCTAP**: Position of the POSTCNT tap on the CYCCNT counter
0: Tap at CYCCNT[6]
1: Tap at CYCCNT[10]

Bits 8:5 **POSTINIT[3:0]**: Initial value of the POSTCNT counter

Writes to this field are ignored if POSTCNT counter is enabled (that is, CYCEVTENA or PCSAMPLENA must be reset prior to writing POSTINIT).

Bits 4:1 **POSTRESET[3:0]**: Reload value of the POSTCNT counter.

Bit 0 **CYCCNTENA**: CYCCNT counter enable

0: Disabled

1: Enabled

DWT cycle count register (M7_DWT_CYCNT)

Address offset: 0x004

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CYCCNT[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYCCNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **CYCCNT[31:0]**: Processor clock cycle counter

DWT CPI count register (M7_DWT_CPICNT)

Address offset: 0x008

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CPICNT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **CPICNT[7:0]**: CPI counter

Counts additional cycles required to execute multi-cycle instructions, except those recorded by DWT_LSUCNT, and counts any instruction fetch stalls.

DWT exception count register (M7_DWT_EXCCNT)

Address offset: 0x00C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EXCCNT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **EXCCNT[7:0]**: Exception overhead cycle counter

Counts the number of cycles spent in exception processing.

DWT sleep count register (M7_DWT_SLPCNT)

Address offset: 0x010

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SLEPCNT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **SLEPCNT[7:0]**: Sleep cycle counter

Counts the number of cycles spent in sleep mode (WFI, WFE, sleep-on-exit).

DWT LSU count register (M7_DWT_LSUCNT)

Address offset: 0x014

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LSUCNT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **LSUCNT[7:0]**: Load store counter

Counts additional cycles required to execute load and store instructions.

DWT fold count register (M7_DWT_FOLDCNT)

Address offset: 0x018

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FOLDCNT[7:0]							
								r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **FOLDCNT[7:0]**: Folded instruction counter

Increments on each instruction that takes 0 cycles.

DWT program counter sample register (M7_DWT_PCSR)

Address offset: 0x01C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EIASAMPLE[31:16]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EIASAMPLE[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:0 **EIASAMPLE[31:0]**: Executed instruction address sample value

Samples the current value of the program counter.

DWT comparator register x (M7_DWT_COMPx)

Address offset: 0x020 + x * 0x10 (for x = 0 to 3)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP[31:16]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **COMP[31:0]**: Reference value for comparison.

DWT mask register x (M7_DWT_MASKx)

Address offset: $0x024 + x * 0x10$ (for $x = 0$ to 3)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MASK[4:0]				
											rw	rw	rw	rw	rw

Bits 31:5 Reserved, must be kept at reset value.

Bits 4:0 **MASK[4:0]**: Comparator mask size

Provides the size of the ignore mask applied to the access address for address range matching by comparator n. A debugger can write 0b11111 to this field and then read the register back to determine the maximum mask size supported.

DWT function register x (M7_DWT_FUNCtx)

Address offset: $0x028 + x * 0x10$ (for $x = 0$ to 3)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	MATCH ED	Res.	Res.	Res.	Res.	DATAVADDR1[3:0]			
							r					rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAVADDR0[3:0]				DATAVSIZE[1:0]		LINK1 ENA	DATAV MATCH	CYC MATCH	Res.	EMIT RANGE	Res.	FUNCTION[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw		rw		rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bit 24 **MATCHED**: Comparator match (read-only)

Indicates if a comparator match has occurred since the register was last read.

0: No match

1: Match occurred

Bits 23:20 Reserved, must be kept at reset value.

- Bits 19:16 **DATAVADDR1[3:0]**: Comparator number of a second comparator
When the DATAVMATCH and LNK1ENA bits are both 1, this field can hold the comparator number of a second comparator to use for linked address comparison.
- Bits 15:12 **DATAVADDR0[3:0]**: Comparator number of a comparator
When the DATAVMATCH and LNK1ENA bits are both 1, this field can hold the comparator number of a comparator to use for linked address comparison.
- Bits 11:10 **DATAVSIZE[1:0]**: Size of required data comparison
For data value matching, specifies the size of the required data comparison.
- 0x0: Byte
0x1: Half word
0x2: Word
0x3: Reserved
- Bit 9 **LNK1ENA**: Support of a second linked comparator (read-only)
Indicates whether the use of a second linked comparator is supported (read-only).
- 1: Supported
- Bit 8 **DATAVMATCH**: Cycle comparison enable
0: Perform address comparison
1: Perform data value comparison
- Bit 7 **CYCMATCH**: Cycle count comparison enable on comparator 0
This field is reserved for other comparators.
- 0: No cycle count comparison
1: Compare DWT_COMP0 with the cycle counter, DWT_CYCCNT
- Bit 6 Reserved, must be kept at reset value.
- Bit 5 **EMITRANGE**: Data trace address offset packet enable
Enables generation of data trace address offset packets (containing data address bits 0 to 15)
0: Disabled
1: Enabled
- Bit 4 Reserved, must be kept at reset value.
- Bits 3:0 **FUNCTION[3:0]**: Action on comparator match
The meaning of this bit field depends on the setting of the DATAVMATCH and CYCMATCH fields. See [\[5\]](#).

DWT CoreSight peripheral identity register 4 (M7_DWT_PIDR4)

Address offset: 0xFD0

Reset value: 0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **4KCOUNT[3:0]**: Register file size

0x0: Register file occupies a single 4 Kbyte region

Bits 3:0 **JEP106CON[3:0]**: JEP106 continuation code

0x4: Arm® JEDEC code

DWT CoreSight peripheral identity register 0 (M7_DWT_PIDR0)

Address offset: 0xFE0

Reset value: 0x0000 0002

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
								r	r						

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PARTNUM[7:0]**: Part number field, bits [7:0]

0x02: DWT part number

DWT CoreSight peripheral identity register 1 (M7_DWT_PIDR1)

Address offset: 0xFE4

Reset value: 0x0000 00B0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **JEP106ID[3:0]**: JEP106 identity code field, bits [3:0]

0xB: Arm® JEDEC code

Bits 3:0 **PARTNUM[11:8]**: Part number field, bits [11:8]

0x0: DWT part number

DWT CoreSight peripheral identity register 2 (M7_DWT_PIDR2)

Address offset: 0xFE8

Reset value: 0x0000 000B

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]		
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVISION[3:0]**: Component revision number

0x0: r0p0

Bit 3 **JEDEC**: JEDEC assigned value

1: Designer ID specified by JEDEC

Bits 2:0 **JEP106ID[6:4]**: JEP106 identity code field, bits [6:4]

0x3: Arm® JEDEC code

DWT CoreSight peripheral identity register 3 (M7_DWT_PIDR3)

Address offset: 0xFEC

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVAND[3:0]**: Metal fix version

0x0: No metal fix

Bits 3:0 **CMOD[3:0]**: Customer modified

0x0: No customer modifications

DWT CoreSight component identity register 0 (M7_DWT_CIDR0)

Address offset: 0xFF0

Reset value: 0x0000 000D

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[7:0]**: Component ID field, bits [7:0]

0x0D: Common ID value

DWT CoreSight component identity register 1 (M7_DWT_CIDR1)

Address offset: 0xFF4

Reset value: 0x0000 00E0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **CLASS[3:0]**: Component ID field, bits [15:12] - component class

0xE: Trace generator component

Bits 3:0 **PREAMBLE[11:8]**: Component ID field, bits [11:8]

0x0: Common ID value

DWT CoreSight component identity register 2 (M7_DWT_CIDR2)

Address offset: 0xFF8

Reset value: 0x0000 0005

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[19:12]**: Component ID field, bits [23:16]

0x05: Common ID value

DWT CoreSight component identity register 3 (M7_DWT_CIDR3)

Address offset: 0xFFC

Reset value: 0x0000 00B1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[27:20]**: Component ID field, bits [31:24]

0xB1: Common ID value

Cortex-M7 DWT register map and reset values

The Cortex-M7 DWT registers are located at address range 0xE0001000 to 0xE0001FFC, on the AHBD.

Table 624. Cortex-M7 DWT register map and reset values

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																		
0x000	M7_DWT_CTRL	NUMCOMP[3:0]				NOTRCPKT		NOEXTTRIG		NOCYCNCNT		NOPRFCNT		Res.		CYCEVTENA		FOLDEVTEANA		LSUEVTENA		SLEEPEVTENA		EXCEVTENA		CPIEVTENA		EXCTRCENA		Res.		Res.		Res.		PCSAMPLENA		SYNCTAP[1:0]		CYCTAP		POSINT[3:0]				POSTRESET[3:0]				CYCCNTENA	
	Reset value	0	1	0	0	0	0	0	0	0		0	0	0	0	0	0	0				0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
0x004	M7_DWT_CYCCNT	CYCCNT[31:0]																																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x008	M7_DWT_CPICNT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CPICNT[7:0]																									
	Reset value																										0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x00C	M7_DWT_EXCCNT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	EXCCNT[7:0]																								
	Reset value																										0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						

Table 624. Cortex-M7 DWT register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x010	M7_DWT_SLPCNT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	SLEEP CNT[7:0]													
	Reset value																									0	0	0	0	0	0	0	0						
0x014	M7_DWT_LSUCNT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	LSUCNT[7:0]													
	Reset value																									0	0	0	0	0	0	0	0						
0x018	M7_DWT_FOLD CNT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	FOLD CNT[7:0]													
	Reset value																									0	0	0	0	0	0	0	0						
0x01C	M7_DWT_PCSR	EIASAMPLE[31:0]																																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x020	M7_DWT_COMP0	COMP[31:0]																																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x024	M7_DWT_MASK0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	MASK[4:0]													
	Reset value																													0	0	0	0	0					
0x028	M7_DWT_FUNC T0	Res	Res	Res	Res	Res	Res	Res	MATCHED	Res	Res	Res	Res	DATAVADDR1[3:0]			DATAVADDR0[3:0]			DATAVSIZE[1:0]			LNK1ENA	DATAVMATCH	CYCMATCH	Res	EMITRANGE	Res	FUNCTION[3:0]										
	Reset value								0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0					
0x02C	Reserved	Reserved																																					
0x030	M7_DWT_COMP1	COMP[31:0]																																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x034	M7_DWT_MASK1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	MASK[4:0]													
	Reset value																												0	0	0	0	0	0					
0x038	M7_DWT_FUNC T1	Res	Res	Res	Res	Res	Res	Res	MATCHED	Res	Res	Res	Res	DATAVADDR1[3:0]			DATAVADDR0[3:0]			DATAVSIZE[1:0]			LNK1ENA	DATAVMATCH	CYCMATCH	Res	EMITRANGE	Res	FUNCTION[3:0]										
	Reset value								0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0					
0x03C	Reserved	Reserved																																					
0x040	M7_DWT_COMP2	COMP[31:0]																																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x044	M7_DWT_MASK2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	MASK[4:0]													
	Reset value																												0	0	0	0	0	0					
0x048	M7_DWT_FUNC T2	Res	Res	Res	Res	Res	Res	Res	MATCHED	Res	Res	Res	Res	DATAVADDR1[3:0]			DATAVADDR0[3:0]			DATAVSIZE[1:0]			LNK1ENA	DATAVMATCH	CYCMATCH	Res	EMITRANGE	Res	FUNCTION[3:0]										
	Reset value								0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0					
0x04C	Reserved	Reserved																																					
0x050	M7_DWT_COMP3	COMP[31:0]																																					
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

Table 624. Cortex-M7 DWT register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x054	M7_DWT_MASK3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MASK[4:0]						
	Reset value																												0	0	0	0	0		
0x058	M7_DWT_FUNC3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MATCHED	Res.	Res.	Res.	Res.	DATAVADDR[3:0]				DATAVADDR0[3:0]				DATAVSIZE[1:0]				LNK1ENA	DATAVMATCH	CYCMATCH	Res.	EMITRANGE	Res.	FUNCTION[3:0]			
	Reset value								0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x05C to 0xFCC	Reserved	Reserved																																	
0xFD0	M7_DWT_PIDR4	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CONF[3:0]					
	Reset value																									0	0	0	0	0	1	0	0		
0xFD4 to 0xFCC	Reserved	Reserved																																	
0xFE0	M7_DWT_PIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]									
	Reset value																									0	0	0	0	0	0	1	0		
0xFE4	M7_DWT_PIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[1:8]					
	Reset value																									1	0	1	1	0	0	0	0		
0xFE8	M7_DWT_PIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]				
	Reset value																									0	0	0	0	1	0	1	1		
0xFEC	M7_DWT_PIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]					
	Reset value																									0	0	0	0	0	0	0	0		
0xFF0	M7_DWT_CIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]									
	Reset value																									0	0	0	0	1	1	0	1		
0xFF4	M7_DWT_CIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]					
	Reset value																									0	1	1	1	0	0	0	0		
0xFF8	M7_DWT_CIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]									
	Reset value																									0	0	0	0	0	1	0	1		
0xFFC	M7_DWT_CIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]									
	Reset value																									1	0	1	1	0	0	0	1		

60.6.3 Cortex-M7 instrumentation trace macrocell (ITM)

The ITM generates trace information as packets. There are four sources that can generate packets. If multiple sources generate packets at the same time, the ITM arbitrates the order in which packets are output. The four sources in decreasing order of priority are:

1. Software trace

Software can write directly to any of 32 x 32-bit ITM stimulus registers to generate packets. The permission level for each port can be programmed. When software writes to an enabled stimulus port, the ITM combines the identity of the port, the size of the write access, and the data written, into a packet that it writes to a FIFO. The ITM